SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Error Analysis Task

Course Code:	CSA17	**Course Title:**	Artificial Intelligence
CO Assessed:	CO1 – Apply (BL3)	**Assessment:**	Assessment Tool 3 – Error Analysis Task
Weightage:	35% of CIA	**Duration / Marks:**	60 mins / 100 marks
CO1 Statement:	Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)		
Student Name:	K Mahesh Sai	**Reg. No.:**	192412674
Section / Batch:	3rd Year	**Date:**	10-10-2026

Code of Conduct

I K Mahesh Sai 192412674 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: K Mahesh Sai

Instructions

- Read each question carefully; answer all four questions (100 in total). Off-Class Assessment Tool.
- Each question gives a student's faulty work. Do not solve from scratch: identify every error (cite it by statement, component, iteration or step number), explain its root cause, correct it with justification and apply the corrected concept. An error merely carried forward counts once.
- Follow the conventions stated in each question (alphabetical expansion, explored set, goal test on removal from the frontier, tie-breaking) and present tables neatly.
- Marks follow the Error Analysis rubric: error identification, root cause analysis, correction and justification, application of concepts, and clarity and presentation.
- Upload in GitHub and GCR (question paper with Reg. No., Name, answers, rubrics and evaluation page at the end).

Section / Topic	Focus	Question Nos.
1. Foundations and History of AI	Four approaches to AI, Turing Test, eras and milestones	**Q1**
2. Intelligent Agents: Agents and Environments	Agent, percepts, actions, agent function and program, PEAS	**Q1**
3. The Concept of Rationality	Performance measure, percept sequence, rational vs omniscient	**Q1**
4. The Nature of Environments	Task-environment properties and their classification	**Q1**
5. The Structure of Agents	Simple reflex and model-based reflex agent programs	**Q1**
6. Problem-Solving Agents	Goal formulation, problem formulation, search, execution	**Q2**
7. Problem Formulation	States, initial state, actions, goal test, path cost; belief states	**Q2, Q4**
8. Searching for Solutions	Search graph, frontier, explored set, node expansion	**Q3**

CSA17 – ARTIFICIAL INTELLIGENCE

Assessment Tool 3 – Error Analysis Task | Complete Solutions

Student: K Mahesh Sai | Reg. No.: 192412674 | Section/Batch: 3rd Year | Date: 10-10-2026

Method: For each item, identify the error, explain the underlying misconception, correct it, and apply the corrected concept. Where a student's later answer follows an earlier error, the originating error is identified once.

QUESTION 1: FOUNDATIONS OF AI, INTELLIGENT AGENTS AND RATIONALITY

(a) Correct / incorrect statements and erroneous claims

Statement	Status	Erroneous phrase / claim
S1	Incorrect	The Turing Test does not test whether internal reasoning follows logic and is not the "thinking rationally" approach. It is an example of acting humanly.
S2	Incorrect	AI winters did not occur in the 1950s before Dartmouth. The major AI winters occurred later, especially in the 1970s and late 1980s/early 1990s. Dartmouth (1956) is associated with naming AI, not reviving it after a 1950s winter.
S3	Incorrect	A rational agent does not know every outcome in advance and is not judged by omniscience. Rationality is based on expected performance given available percepts, knowledge, and actions.
S4	Incorrect	Performance measure and actuators are swapped. Performance describes success criteria; actuators are mechanisms/actions used to affect the environment.
S5	Incorrect	Without a clock, chess is fully observable, deterministic, sequential, static, and discrete —not partially observable, stochastic, episodic.
S6	Correct	This correctly describes a model-based reflex agent's internal state and use of a model of world evolution and action effects.

(b) Root-cause analysis

Error	Root cause / misconception
S1	Confuses the Turing Test (human-like observable behavior) with the logical-rationality approach (reasoning according to formal rules).
S2	Confuses the 1956 Dartmouth milestone with later funding and interest downturns known as AI winters.
S3	Confuses rationality with omniscience. Rational choices maximize expected performance using information actually available; outcomes can remain uncertain.
S4	Does not distinguish a performance measure (what counts as success) from actuators (how the agent acts).
S5	Misclassifies standard chess properties. The board and all pieces are visible; legal moves have predictable outcomes; moves affect later states; no clock means the environment does not change while the agent deliberates.
S6	No error: the statement correctly identifies internal state and a world model.

(c) Corrected statements and justification

Statement	Corrected statement	Justification
S1	The Turing Test (1950) evaluates whether a machine's conversational behavior is indistinguishable from a human's; it is an example of the acting-humanly approach.	It evaluates outward behavior, not whether internal reasoning follows formal logic.

S2	The Dartmouth workshop in 1956 is associated with the naming of artificial intelligence. AI winters were later periods of reduced funding and interest, notably in the 1970s and late 1980s/early 1990s.	The historical order in the original note is incorrect.
S3	A rational agent selects the action expected to maximize its performance measure based on its percept sequence, prior knowledge, available actions, and performance measure. It need not predict every outcome.	Rationality is bounded by available information; omniscience means knowing actual outcomes in advance.
S4	PEAS: Performance measure—safe, fast, legal, comfortable trip. Environment—roads, other traffic, pedestrians, weather. Actuators—steering, accelerator, brake, horn. Sensors—cameras, GPS, speedometer.	This assigns each PEAS component to its proper role.
S5	Chess without a clock is fully observable, multi-agent, deterministic, sequential, static, and discrete.	The complete board is visible; two players compete; moves have predictable results; moves affect future states; the board does not change while deliberating; states/actions are discrete.
S6	No change.	A model-based reflex agent maintains and updates internal state using a model of how the world changes and how actions affect it.

(d) Application: self-driving taxi and unforeseeable landslide

(i) Was the taxi's choice rational?

Yes, assuming the taxi selected the highway because all available traffic reports indicated it was clear and this choice maximized expected performance at the time. The four factors are: (1) performance measure—safe, timely, legal, comfortable travel; (2) prior knowledge—the taxi's knowledge of roads and traffic; (3) available actions—possible routes and driving actions; and (4) percept sequence—the reports and sensor readings available before choosing. An unforeseeable landslide is a bad outcome, but it does not by itself prove the earlier decision irrational.

(ii) Task-environment properties that prevent perfect foresight

The taxi environment is partially observable (not every hazard is visible or reported), stochastic (unexpected events can occur), dynamic (road conditions change), sequential (current route choices affect later states), multi-agent (other road users act independently), and partly continuous (position, speed, and steering vary continuously). These properties make complete prediction impossible.

(iii) A simple-reflex failure and model-based improvement

Example: after cleaning or encountering a clear patch of road, a simple reflex taxi might repeatedly choose the same action from the current percept and fail to remember that a road segment was reported blocked. A model-based reflex taxi maintains an internal state, such as a remembered map of recent road conditions, its location, and the effects of previous actions, and updates it as new percepts arrive.

QUESTION 2: PROBLEM-SOLVING AGENTS AND PROBLEM FORMULATION

(a) Check each component and note

Component / note	Status	What is wrong
State representation	Incorrect	(M,C) omits the boat's bank. The same counts with the boat on different banks have different legal actions.
Initial state	Incorrect / incomplete	Must include boat location: (3,3,L).
Actions	Mostly correct, but incomplete as a formulation	The listed passenger loads are the possible loads for a boat of capacity 1 or 2, but each action must also specify the boat's direction/side and be applicable only when those passengers and the boat are available on that bank.
Goal test	Incorrect	M=C is not the goal. The goal is all missionaries and cannibals on the right bank, with the boat on the right: (0,0,R).
Legality check	Incorrect	Must check safety on both banks: on each

		bank, missionaries are not outnumbered by cannibals unless there are zero missionaries there.
Path cost	Incorrect	The objective is fewest crossings, so each crossing costs 1, regardless of whether one or two people are carried.
Note N1	Incorrect	Correct order: goal formulation → problem formulation → search → execution.
Note N2	Incorrect / ambiguous	A full syntactic state representation has 4×4×2 = 32 combinations. Ten (M,C) pairs satisfy the safety rule on both banks, giving 20 safe states when boat side is included; 16 of those safe states are reachable from the specified initial state. The note mixes up count-pairs with full states/reachability.

(b) Root-cause analysis

Error	Root cause
State / initial state	A state must contain all information needed to determine available actions and future transitions, including the boat side.
Actions	Actions were written only as passenger combinations without explicitly accounting for boat direction, availability, and legality.
Goal test	Confuses a numerical relation M=C with the actual desired world configuration.
Legality check	Checks only the left bank and ignores that the right bank can also violate the safety constraint.
Path cost	Optimizes people transported instead of the specified objective, number of crossings.
N1	Reverses the logical order: the agent must define what it wants, formulate the problem, search, and then execute.
N2	Counts only (M,C) pairs, omits the boat side, and confuses all possible states with safe and reachable states.

(c) Corrected formulation

Component	Corrected formulation	Justification
State	(M,C,B), where M and C are counts on the left bank and $B \in \{L,R\}$ is the boat side.	Tracks both population distribution and boat location.
Initial state	(3,3,L).	All six people and the boat begin on the left.
Actions	Move one missionary (1M), two missionaries (2M), one cannibal (1C), two cannibals (2C), or one of each (1M+1C) across the river in the boat's current direction, when available and safe.	These are the possible one- or two-person loads; transitions must obey capacity, availability and safety.
Transition model	Subtract the passengers from (M,C) when crossing L→R; add them when crossing R→L; flip B. Reject out-of-range or unsafe resulting states.	Defines exactly how each legal crossing changes the state.
Goal test	(M,C,B)=(0,0,R).	Everyone is on the right bank and the boat is there too.
Legality	For each bank, either missionaries = 0 or missionaries ≥ cannibals.	The rule applies to both banks.
Path cost	1 per crossing; total path cost equals the number of crossings.	Matches "fewest crossings."
Agent phases	Goal formulation → problem formulation → search → execution.	The agent defines the goal, formalizes the problem, finds a plan, then performs it.

(d)(i) Successors from the initial state (3,3,L)

Boat load	Resulting state (M,C,B)	Legal? Reason
1M	(2,3,R)	No. Left bank has 2 missionaries and 3 cannibals, so missionaries are outnumbered.
2M	(1,3,R)	No. Left bank has 1 missionary and 3 cannibals.
1C	(3,2,R)	Yes. Left bank 3M≥2C; right bank has 0 missionaries, so it is safe.
2C	(3,1,R)	Yes. Both banks satisfy the safety rule.
1M + 1C	(2,2,R)	Yes. Left bank has equal numbers; right bank has 2M and 2C.

(d)(ii) Example of a state incorrectly accepted by the student's check

State (2,0,L) passes the student's left-bank check because 0 cannibals do not outnumber 2 missionaries. However, on the right bank there are 1 missionary and 3 cannibals, so the state is actually illegal. This demonstrates why both banks must be checked.

(d)(iii) Optimal path cost

An optimal solution requires 11 crossings, so the path cost is 11.

Crossing	Move	State after crossing (M,C,B)
0	Start	(3,3,L)
1	2C →	(3,1,R)
2	1C ←	(3,2,L)
3	2C →	(3,0,R)
4	1C ←	(3,1,L)
5	2M →	(1,1,R)
6	1M+1C ←	(2,2,L)
7	2M →	(0,2,R)
8	1C ←	(0,3,L)
9	2C →	(0,1,R)
10	1C ←	(0,2,L)
11	2C →	(0,0,R)

QUESTION 3: SEARCHING FOR SOLUTIONS AND UNINFORMED SEARCH

Graph edges: S–C(3), S–D(3), A–B(2), A–C(6), B–D(4), B–G(3), D–E(2), D–G(6), E–G(3). Neighbours are expanded alphabetically; goal test occurs when the goal is removed from the frontier.

(a) Identify errors in the traces and remarks

Where	Error
BFS iteration 3	Queue should be [A, B, E, G], not [B, E, G, A]. A was already in the queue and must remain ahead of newly appended neighbours.
BFS iterations 4–6 / result	The later trace is carried forward from the incorrect queue order and omits the correct final dequeue of G at iteration 7. The BFS path S–D–G and cost 9 are correct, but the stated 6 nodes removed is wrong; it is 7.
DFS iteration 1	Stack should be [C,D] (top at left) so alphabetically first C is popped first, not [D,C].
DFS iterations 2–3 / result	The trace follows the wrong stack order. Correct DFS visits S,C,A,B,G; path is S–C–A–B–G with cost 14 and 5 nodes removed.
UCS iteration 4	Expanding E at cost 5 discovers G at cost 8, improving the frontier entry G:9(D). It must be replaced with G:8(E).
UCS iterations 5–7 / result	These rows carry forward the missed decrease-key update. Correct UCS removes G at cost 8 after S,C,D,E,B; total nodes removed = 6.
Remark R1	False: graph search needs an explored set to avoid repeated expansion and cycles.
Remark R2	False: equal returned paths/costs do not make every trace correct or every algorithm optimal. Here BFS finds cost 9, DFS cost 14, UCS cost 8.
Remark R3	False: fewest nodes removed in this instance does not make DFS generally best; DFS is not guaranteed to find a shortest or least-cost path.

(b) Root-cause analysis

Error	Root cause
BFS order	Queue is FIFO; previously queued A cannot be moved behind new successors.
BFS count	The goal test is on removal from the frontier; the correct queue order requires one additional dequeue before G.
DFS stack	Stack is LIFO; to pop the alphabetically first neighbour first, child

	insertion order must be chosen accordingly.
DFS conclusion	DFS follows traversal order, not shortest path cost; its first goal path may have more edges/cost.
UCS update	UCS orders frontier by cumulative path cost and must replace a frontier node's entry when a cheaper route is found.
UCS carry-forward	The failure to update G's cost leads to an incorrect goal cost and node-removal trace.
R1–R3	Confuses an algorithm's performance on one graph with general guarantees, and confuses BFS's edge-depth optimality with cost optimality.

(c)(i) Corrected BFS

Iteration	Node dequeued	Queue after expansion	Explored set
1	S	[C, D]	{S}
2	C	[D, A]	{S, C}
3	D	[A, B, E, G]	{S, C, D}
4	A	[B, E, G]	{S, C, D, A}
5	B	[E, G]	{S, C, D, A, B}
6	E	[G]	{S, C, D, A, B, E}
7	G (goal)	[]	{S, C, D, A, B, E}

Queue after iteration 3: [A, B, E, G]. Dequeue order: S, C, D, A, B, E, G. Path found: S–D–G; path cost = 3 + 6 = 9.

(c)(ii) Corrected DFS

Iteration	Node popped	Stack after pushing children (top at left)
1	S	[C, D]
2	C	[A, D]
3	A	[B, D]
4	B	[G, D]
5	G (goal)	[D]

Path found: S–C–A–B–G. Path cost = 3 + 6 + 2 + 3 = 14. Nodes removed from frontier = 5.

(c)(iii) Corrected uniform-cost search

Iteration	Node expanded	g(n)	Priority queue after expansion (lowest cost first)
1	S	0	C:3(S), D:3(S)
2	C	3	D:3(S), A:9(C)
3	D	3	E:5(D), B:7(D), A:9(C), G:9(D)
4	E	5	B:7(D), G:8(E), A:9(C)
5	B	7	G:8(E), A:9(C)
6	G (goal)	8	—

UCS path: S–D–E–G. Path cost = 3 + 2 + 3 = 8. Nodes removed from frontier = 6.

(c) Justification of corrections

BFS preserves FIFO order and therefore returns a path with the fewest edges, but not necessarily the lowest weighted cost. DFS uses a LIFO stack and may find a longer or more expensive path. UCS expands the lowest cumulative-cost frontier node; when E reveals a cheaper route to G (cost 8), G's old cost-9 entry must be replaced.

(d)(i) Comparison

Strategy	Path found	Path cost	Nodes removed	Optimal here?
BFS	S–D–G	9	7	Fewest edges; not least weighted cost
DFS	S–C–A–B–G	14	5	No
UCS	S–D–E–G	8	6	Yes, least-cost path

(d)(ii) Why BFS needs an explored set

At iteration 3, D's neighbours include S, which has already been explored, and B, E, G, while A is already waiting in the queue. The explored set prevents returning to S and expanding it again around the undirected cycle S–D–S. The queue check also prevents duplicate entries such as A.

QUESTION 4: SEARCH WITH PARTIAL INFORMATION – BELIEF STATES

State key: S1=(L,D,D), S2=(R,D,D), S3=(L,D,C), S4=(R,D,C), S5=(L,C,D), S6=(R,C,D), S7=(L,C,C), S8=(R,C,C).

(a) Errors in Task A, Task B and conclusions

Where	Error
Task A, step 3 (Suck)	From {S6,S8}, Suck at the right square makes both possible states S8. Correct result is {S8}, not {S6,S8}.
Task B, start percept update	Percept [Left, Dirty] should filter all eight states to {S1,S3}; student's {S1,S2,S3,S6} includes states at the wrong location or with a clean left square.
Task B, step 1 prediction	Suck from {S1,S3} predicts {S5,S7}, not {S4,S5,S7,S8}.
Task B, step 2 prediction	Right from {S5,S7} predicts {S6,S8}, not {S6,S7}. Right changes the agent's location to R.
Conclusion C1	Incorrect: after the corrected sensorless plan the belief state is a singleton, and sensorless problems can be solved when a plan guarantees the goal for every possible state.
Conclusion C2	The singleton {S6} identifies the state, but S6=(R,C,D) still has the right square dirty; the goal that both squares are clean has not been reached.
Conclusion C3	There are 2^8 = 256 subsets of eight physical states (255 non-empty subsets). Belief-state search can therefore have many more states than physical-state search.

(b) Root-cause analysis

Error	Root cause
Task A step 3	The action transition was not applied to every state in the belief set. Suck cleans the current square; it cannot leave S6 unchanged because its right square is dirty.
Task B initial update	A percept must remove all states inconsistent with the observed location and current-square status.
Task B step 1	Prediction is the union of successors of each state after applying the action; it is not a percept filter.
Task B step 2	Right changes the agent location from L to R for both candidate states.
C1	Confuses uncertainty about the exact physical state with inability to guarantee a goal; sensorless plans can achieve a goal across all states in a belief set.
C2	Confuses knowing the exact state with satisfying the goal test.
C3	Confuses the number of physical states with the number of subsets of those states.

(c)(i) Corrected Task A: sensorless plan [Suck, Right, Suck]

Step	Action	Belief state after action
0	Start	{S1, S3, S5, S7}
1	Suck	{S5, S7}
2	Right	{S6, S8}
3	Suck	{S8}

The plan succeeds: the final belief state is {S8}, and S8=(R,C,C), so both squares are clean.

(c)(ii) Corrected Task B: percept-based updates

Step	Action	After action (predicted)	Percept	After percept (updated)
0	Start	{S1,S2,S3,S4,S5,S6,S7,S8}	[Left, Dirty]	{S1,S3}
1	Suck	{S5,S7}	[Left, Clean]	{S5,S7}
2	Right	{S6,S8}	[Right, Dirty]	{S6}

(c)(iii) Corrected conclusions

C1 corrected: The sensorless plan [Suck, Right, Suck] takes the initial belief state {S1,S3,S5,S7} to {S8}; it guarantees both squares clean even though the agent initially lacks sensors. The exact location need not be known at every step for a plan to guarantee the goal.

C2 corrected: After the percept [Right, Dirty], the updated belief state is {S6}, so the agent knows it is in state S6=(R,C,D). It knows the left square is clean and the right square is dirty; therefore the goal has not been reached.

(d)(i) Number of belief states

With 8 physical states, there are 2^8 = 256 possible subsets, including the empty set; if only non-empty beliefs are considered, there are 255. This can make belief-state search much harder because the search may need to consider subsets of physical states rather than only individual states.

(d)(ii) Apply plan [Right, Suck] to {S1,S3,S5,S7}

Step	Action	Belief state after action
0	Start	{S1,S3,S5,S7}
1	Right	{S2,S4,S6,S8}
2	Suck	{S4,S8}

Solution? No. The final belief state {S4,S8} includes S4=(R,D,C), where the left square is still dirty. Since every possible state in a goal belief must satisfy both squares clean, this plan does not guarantee the goal.

FINAL ANSWER SUMMARY

Question	Key results
Q1	S1–S5 incorrect; S6 correct. PEAS roles corrected. Rationality is judged from available information, not omniscience.
Q2	State=(M,C,boat side); initial (3,3,L); goal (0,0,R); safety checked on both banks; cost=1 per crossing; optimal cost=11 crossings.
Q3	BFS path S–D–G, cost 9, 7 removals. DFS path S–C–A–B–G, cost 14, 5 removals. UCS path S–D–E–G, cost 8, 6 removals; UCS is least-cost optimal.
Q4	Task A corrected final belief {S8}; Task B updates {S1,S3}→{S5,S7} →{S6}; 256 belief subsets (255 non-empty). Plan [Right,Suck] ends in {S4,S8}, not a guaranteed solution.

End of complete solutions.